

Newman-Penrose Formalism Calculations

Stephani et al., Equation (28.56b_e1)

Metric

$$ds^2 = \left(\frac{8\kappa_0 q_0^2 r e^x - (\kappa_0 q_0^2 e^x - 4m_0 r)u}{2r^2 u} \right) du \otimes du - du \otimes dr - dr \otimes du + r^2 dx \otimes dx + r^2 dy \otimes dy$$

Coordinates

$$(x^\mu) = (u, r, x, y)$$

Null Tetrad

Covariant components

$$l = (1) du$$

$$n = \left(\frac{\kappa_0 q_0^2 e^x}{4r^2} - \frac{2\kappa_0 q_0^2 e^x}{ru} - \frac{m_0}{r} \right) du + (1) dr$$

$$m = \left(\frac{1}{2} \sqrt{2} r \right) dx + \left(-\frac{1}{2} i \sqrt{2} r \right) dy$$

$$\bar{m} = \left(\frac{1}{2} \sqrt{2} r \right) dx + \left(\frac{1}{2} i \sqrt{2} r \right) dy$$

Contravariant components

$$l = (-1) dr$$

$$n = (-1) \, du + \left(\frac{\kappa_0 q_0^2 e^x}{4 r^2} - \frac{2 \kappa_0 q_0^2 e^x}{r u} - \frac{m_0}{r} \right) dr$$

$$m = \left(\frac{\sqrt{2}}{2 r} \right) dx + \left(-\frac{i \sqrt{2}}{2 r} \right) dy$$

$$\overline{m} = \left(\frac{\sqrt{2}}{2 r} \right) dx + \left(\frac{i \sqrt{2}}{2 r} \right) dy$$

Directional Derivatives

$$D = -\partial_r$$

$$\Delta = -\partial_u + \frac{\kappa_0 q_0^2 e^x}{4 r^2} - \frac{2 \kappa_0 q_0^2 e^x}{r u} - \frac{m_0}{r} \partial_r$$

$$\delta = \frac{\sqrt{2}}{2 r} \partial_x - \frac{i \sqrt{2}}{2 r} \partial_y$$

$$\bar{\delta} = \frac{\sqrt{2}}{2 r} \partial_x + \frac{i \sqrt{2}}{2 r} \partial_y$$

Spin Coefficients

$$\rho = \frac{1}{r}$$

$$\sigma = 0$$

$$\kappa = 0$$

$$\mu = \frac{\kappa_0 q_0^2 e^x}{4 r^3} - \frac{2 \kappa_0 q_0^2 e^x}{r^2 u} - \frac{m_0}{r^2}$$

$$\lambda = 0$$

$$\nu = \frac{\sqrt{2} \kappa_0 q_0^2 e^x}{8 r^3} - \frac{\sqrt{2} \kappa_0 q_0^2 e^x}{r^2 u}$$

$$\alpha = 0$$

$$\beta = 0$$

$$\tau = 0$$

$$\pi = 0$$

$$\epsilon = 0$$

$$\gamma = \frac{\kappa_0 q_0^2 e^x}{4 r^3} - \frac{\kappa_0 q_0^2 e^x}{r^2 u} - \frac{m_0}{2 r^2}$$

Ricci Tensor Components

$$\Phi_{00} = 0$$

$$\Phi_{01} = 0$$

$$\Phi_{02} = 0$$

$$\Phi_{10} = 0$$

$$\Phi_{11} = \frac{\kappa_0 q_0^2 e^x}{4 r^4}$$

$$\Phi_{12} = \frac{\sqrt{2} \kappa_0 q_0^2 e^x}{8 r^4} - \frac{\sqrt{2} \kappa_0 q_0^2 e^x}{2 r^3 u}$$

$$\Phi_{20} = 0$$

$$\Phi_{21} = \frac{\sqrt{2} \kappa_0 q_0^2 e^x}{8 r^4} - \frac{\sqrt{2} \kappa_0 q_0^2 e^x}{2 r^3 u}$$

$$\Phi_{22} = \frac{\kappa_0 q_0^2 e^x}{8 r^4} + \frac{2 \kappa_0 q_0^2 e^x}{r^2 u^2} - \frac{\kappa_0 q_0^2 e^x}{r^3 u}$$

$$\Lambda = 0$$

Weyl Tensor Components

$$\Psi_0 = 0$$

$$\Psi_1 = 0$$

$$\Psi_2 = \frac{\kappa_0 q_0^2 e^x}{2 r^4} - \frac{2 \kappa_0 q_0^2 e^x}{r^3 u} - \frac{m_0}{r^3}$$

$$\Psi_3 = \frac{\sqrt{2} \kappa_0 q_0^2 e^x}{4 r^4} - \frac{3 \sqrt{2} \kappa_0 q_0^2 e^x}{2 r^3 u}$$

$$\Psi_4 = \frac{\kappa_0 q_0^2 e^x}{8 r^4} - \frac{\kappa_0 q_0^2 e^x}{r^3 u}$$

Newman-Penrose Equations

Equations are vanished.

Bianchi Identities

Equations are vanished.

Petrov Type

Type "II"